

Fun with Damped Harmonic Oscillation

Solutions

1 A Rocking Jar

Basic idea behind underdamped oscillations

When the jar is given a small nudge, it oscillates back and forth like a harmonic oscillator. Because of air resistance and friction, energy is continuously lost, so the oscillations do not persist forever. In the **underdamped** case, the system still oscillates, but its amplitude decays exponentially with time.

A simple linear damping model assumes the damping force is proportional to velocity:

$$F_d \propto -\dot{x}.$$

The equation of motion becomes

$$\ddot{x} + \gamma \dot{x} + \omega_0^2 x = 0,$$

where γ is the linear damping coefficient and ω_0 is the natural angular frequency.

For the underdamped condition,

$$\gamma < 2\omega_0,$$

the displacement solution is

$$x(t) = Ae^{-\gamma t/2} \cos(\omega_u t + \phi),$$

where the damped angular frequency is

$$\omega_u = \sqrt{\omega_0^2 - \frac{\gamma^2}{4}}.$$

For an underdamped oscillator, the angular displacement can be written as

$$\theta(t) = \theta_0 e^{-\gamma t/2} \cos(\omega_u t + \phi).$$

The measured quantity in the rocking jar experiment is the angular velocity

$$\omega(t) = \dot{\theta}(t).$$

Differentiate $\theta(t)$:

$$\begin{aligned} \omega(t) &= \frac{d}{dt} \left[\theta_0 e^{-\gamma t/2} \cos(\omega_u t + \phi) \right] \\ \Rightarrow \omega(t) &= \theta_0 \left(-\frac{\gamma}{2} e^{-\gamma t/2} \cos(\omega_u t + \phi) - \omega_u e^{-\gamma t/2} \sin(\omega_u t + \phi) \right) \\ \omega(t) &= -\theta_0 e^{-\gamma t/2} \left[\frac{\gamma}{2} \cos(\omega_u t + \phi) + \omega_u \sin(\omega_u t + \phi) \right]. \end{aligned}$$

The term in brackets is just a sinusoidal function with the same frequency ω_u , so it can always be rewritten as a single cosine with a shifted phase:

$$\omega(t) = \omega_{\max} e^{-\gamma t/2} \cos(\omega_u t + \tilde{\phi})$$

for some constant $\omega_{\max}$ and phase $\tilde{\phi}$.

Hence the peak (envelope) of angular velocity still decays exponentially:

$$|\omega_{\max}(t)| \propto e^{-\gamma t/2}$$

How to obtain the time period T and ω_u from the data

The data recorded is angular velocity $\omega(t)$ as a function of time t . The signal $\omega(t)$ oscillates, so we can measure the period T directly from the time difference between two successive peaks (two maxima or two minima).

If the peak times are t_1 and t_2 , then

$$T = t_2 - t_1.$$

For better accuracy, average over many cycles:

$$T_{\text{avg}} = \frac{t_{n+N} - t_n}{N}.$$

Then the oscillation frequency is

$$\omega_u = \frac{2\pi}{T_{\text{avg}}}.$$

How to obtain γ from decay of peak amplitudes

Even though the recorded quantity is angular velocity $\omega(t)$, its peak magnitudes follow the same exponential decay envelope:

$$|\omega_{\text{max}}(t)| = A_0 e^{-\gamma t/2}.$$

Take two successive peak magnitudes $A_1 = |\omega_{\text{max}}(t_1)|$ and $A_2 = |\omega_{\text{max}}(t_2)|$. Then

$$\frac{A_2}{A_1} = e^{-\gamma(t_2 - t_1)/2}.$$

Taking the natural logarithm gives

$$\ln\left(\frac{A_1}{A_2}\right) = \frac{\gamma}{2}(t_2 - t_1),$$

so

$$\gamma = \frac{2}{t_2 - t_1} \ln\left(\frac{A_1}{A_2}\right).$$

Once ω_u and γ are obtained from the data, the natural frequency is

$$\omega_0 = \sqrt{\omega_u^2 + \frac{\gamma^2}{4}}$$

(e) Plotting analytical solution vs data

We plot the measured $\omega_{\text{data}}(t)$ and the fitted $\omega_{\text{fit}}(t)$ on the same graph. The peak values should follow an exponential decay envelope:

$$|\omega_{\text{max}}(t)| = A_0 e^{-\gamma t/2}.$$

(f) linear damping is not perfect

In reality, the decay of oscillations may deviate from this simple behavior as the model assumes damping is proportional to velocity ($F_d \propto -\dot{x}$), giving exponential decay. In real experiments, damping may be affected by nonlinear friction, surface roughness, and air drag, so the decay is not perfectly exponential. A better test is to repeat the experiment on different surfaces (smooth vs rough) and compare the damping behavior.

2 Gotham Needs You!

The 1-D equation of motion of a body of mass m undergoing damped oscillations (with the resistive force proportional to its velocity) is given by

$$m \frac{d^2 x}{dt^2} + b \frac{dx}{dt} + kx = 0, \quad (1)$$

or

$$\frac{d^2 x}{dt^2} + \gamma \frac{dx}{dt} + \omega_0^2 x = 0, \quad (2)$$

where $\gamma = b/m$ is the damping coefficient and $\omega_0 = \sqrt{k/m}$ is the natural frequency of oscillation.

(a) Assume a trial solution of the form

$$x(t) = Ce^{\alpha t} \quad (3)$$

substituting into the equation of motion,

$$\alpha^2 Ce^{\alpha t} + \gamma \alpha Ce^{\alpha t} + \omega_0^2 Ce^{\alpha t} = 0 \quad (4)$$

Since $Ce^{\alpha t} \neq 0$,

$$\alpha^2 + \gamma \alpha + \omega_0^2 = 0 \quad (5)$$

This is the characteristic equation. The roots are

$$\alpha = \frac{-\gamma \pm \sqrt{\gamma^2 - 4\omega_0^2}}{2} \quad (6)$$

The nature of motion depends on the term $\gamma^2 - 4\omega_0^2$

Case (i) Underdamping, $\gamma < 2\omega_0$

$$\alpha = \frac{-\gamma \pm i\sqrt{4\omega_0^2 - \gamma^2}}{2} \quad (7)$$

let

$$\omega_u = \sqrt{\omega_0^2 - \frac{\gamma^2}{4}} \quad (8)$$

therefore,

$$\alpha = -\frac{\gamma}{2} \pm i\omega_u \quad (9)$$

The general solution is

$$x(t) = C_1 e^{(-\frac{\gamma}{2} + i\omega_u)t} + C_2 e^{(-\frac{\gamma}{2} - i\omega_u)t} \quad (10)$$

Writing the solution in real form,

$$x(t) = Ae^{-\gamma t/2} \cos(\omega_u t + \phi) \quad (11)$$

Case (ii) overdamping, $\gamma > 2\omega_0$

$$\alpha_{1,2} = -\frac{\gamma}{2} \pm \sqrt{\frac{\gamma^2}{4} - \omega_0^2} \quad (12)$$

the roots are real and distinct.

Define

$$\mu_1 = \frac{\gamma}{2} + \sqrt{\frac{\gamma^2}{4} - \omega_0^2}, \quad \mu_2 = \frac{\gamma}{2} - \sqrt{\frac{\gamma^2}{4} - \omega_0^2} \quad (13)$$

with $\mu_1 > \mu_2 > 0$.

The general solution is

$$x(t) = C_1 e^{-\mu_1 t} + C_2 e^{-\mu_2 t} \quad (14)$$

Case(iii) Critical damping, $\gamma = 2\omega_0$,

$$\alpha = -\gamma/2 \quad (15)$$

the roots are repeated. The general solution in this case is written as

$$x(t) = (C_1 + C_2 t)e^{-\omega_0 t} \quad (16)$$

(b) As the overdamped solution is ,

$$x_{\text{overdamped}}(t) = C_1 \exp(-\mu_1 t) + C_2 \exp(-\mu_2 t), \quad (17)$$

with $\mu_1 > \mu_2 > 0$.

μ_1 and μ_2 in terms of γ and ω_0 is written as

$$\mu_1 = \frac{\gamma}{2} + \sqrt{\frac{\gamma^2}{4} - \omega_0^2}, \quad \mu_2 = \frac{\gamma}{2} - \sqrt{\frac{\gamma^2}{4} - \omega_0^2} \quad (18)$$

here, we can see $\mu_1 > \mu_2$, therefore $\exp(-\mu_1 t)$ term decays faster whereas $\exp(-\mu_2 t)$ terms decays slower. So, for the late-time behavior, $t \rightarrow \infty$, the later term dominates and,

$$x(t) \sim C_2 e^{-\mu_2 t}.$$

Thus, the slower-decaying exponential (the term with μ_2) controls the late-time relaxation of the overdamped oscillator.

(c) For critical damping, the damping coefficient satisfies

$$\gamma = 2\omega_0.$$

The equation of motion becomes

$$\ddot{x} + \gamma \dot{x} + \omega_0^2 x = 0.$$

Using $\omega_0^2 = \gamma^2/4$, this can be written as

$$\ddot{x} + \gamma \dot{x} + \frac{\gamma^2}{4} x = 0.$$

Consider the solution

$$x(t) = C t e^{-\gamma t/2},$$

where C is a constant.

First derivative:

$$\dot{x}(t) = C e^{-\gamma t/2} - \frac{\gamma}{2} C t e^{-\gamma t/2} = C e^{-\gamma t/2} \left(1 - \frac{\gamma t}{2}\right).$$

Second derivative:

$$\ddot{x}(t) = -\frac{\gamma}{2} C e^{-\gamma t/2} \left(1 - \frac{\gamma t}{2}\right) - \frac{\gamma}{2} C e^{-\gamma t/2} = C e^{-\gamma t/2} \left(\frac{\gamma^2 t}{4} - \gamma\right).$$

Substitute $x(t)$, $\dot{x}(t)$, and $\ddot{x}(t)$ into the equation of motion:

$$\ddot{x} + \gamma \dot{x} + \frac{\gamma^2}{4} x = 0.$$

This gives

$$C e^{-\gamma t/2} \left(\frac{\gamma^2 t}{4} - \gamma\right) + \gamma C e^{-\gamma t/2} \left(1 - \frac{\gamma t}{2}\right) + \frac{\gamma^2}{4} C t e^{-\gamma t/2} = 0$$

$$C e^{-\gamma t/2} \left[\frac{\gamma^2 t}{4} - \gamma + \gamma - \frac{\gamma^2 t}{2} + \frac{\gamma^2 t}{4}\right] = 0$$

Simplifying the bracketed terms,

$$C e^{-\gamma t/2} (0) = 0.$$

Hence,

$$x(t) = C t e^{-\gamma t/2}$$

is indeed a solution of the differential equation under critical damping.

Since $e^{-\gamma t/2}$ is also a solution, the most general solution for critical damping is

$$x_{\text{crit}}(t) = (C_1 + C_2 t) e^{-\gamma t/2}.$$

- (d) We compare the late-time behavior of the critically damped and overdamped oscillators for the same initial conditions.

For critical damping,

$$\gamma = 2\omega_0,$$

the displacement is given by

$$x_{\text{crit}}(t) = (C_1 + C_2 t) e^{-\gamma t/2}.$$

As $t \rightarrow \infty$, the polynomial factor grows only linearly in time, while the exponential term decays exponentially. Hence, the late-time behavior is dominated by

$$x_{\text{crit}}(t) \sim t e^{-\gamma t/2}.$$

Thus, the decay rate in the critically damped case is governed by the exponential factor

$$e^{-(\gamma/2)t}.$$

For overdamping ($\gamma > 2\omega_0$), the solution is

$$x_{\text{over}}(t) = C_1 e^{-\mu_1 t} + C_2 e^{-\mu_2 t}, \quad \mu_1 > \mu_2 > 0.$$

As $t \rightarrow \infty$, the faster-decaying term $e^{-\mu_1 t}$ becomes negligible, and the motion is dominated by the slower mode:

$$x_{\text{over}}(t) \sim C_2 e^{-\mu_2 t}.$$

The slow decay constant is

$$\mu_2 = \frac{\gamma}{2} - \sqrt{\frac{\gamma^2}{4} - \omega_0^2}.$$

Since $\gamma > 2\omega_0$, it follows that

$$\mu_2 < \frac{\gamma}{2}.$$

The late-time decay rates are therefore:

$$x_{\text{crit}}(t) \sim e^{-(\gamma/2)t}, \quad x_{\text{over}}(t) \sim e^{-\mu_2 t}.$$

Because

$$\frac{\gamma}{2} > \mu_2,$$

the exponential decay in the critically damped case is faster than in the overdamped case.

Therefore, the critically damped oscillator returns to equilibrium faster than the overdamped oscillator.

- (e) We now compare the late-time behavior of the critically damped and underdamped oscillators for the same initial conditions. In the underdamped case, the amplitude envelope decays as

$$A_{\text{under}}(t) \propto e^{-\gamma t/2}, \quad \text{with } \gamma < 2\omega_0.$$

In the critically damped case, $\gamma = 2\omega_0$, and the dominant exponential decay scale is

$$A_{\text{crit}}(t) \propto e^{-\omega_0 t}.$$

Now compare the exponents. Since $\gamma < 2\omega_0$,

$$\frac{\gamma}{2} < \omega_0.$$

Therefore, for $t > 0$,

$$e^{-\omega_0 t} < e^{-(\gamma/2)t}.$$

Therefore, critical damping decays faster than underdamping because of the different decay rate of the oscillations.

3 Still Time?

For a lightly damped pendulum, the oscillation amplitude decays as

$$A(t) = A_0 e^{-\gamma t/2},$$

where γ is the linear damping coefficient.

The time $t_{1/2}$ at which the amplitude reduces to half its initial value is defined by

$$A(t_{1/2}) = \frac{1}{2} A_0.$$

Substituting into the amplitude decay expression,

$$\frac{1}{2} = e^{-\gamma t_{1/2}/2}.$$

Taking the natural logarithm of both sides,

$$-\frac{\gamma t_{1/2}}{2} = \ln\left(\frac{1}{2}\right) = -\ln 2.$$

Hence,

$$\gamma = \frac{2 \ln 2}{t_{1/2}}.$$

For pendulum A ,

$$t_{1/2,A} = 100 \text{ s} \quad \Rightarrow \quad \gamma_A = \frac{2 \ln 2}{100}.$$

For pendulum B ,

$$t_{1/2,B} = 1000 \text{ s} \quad \Rightarrow \quad \gamma_B = \frac{2 \ln 2}{1000}.$$

Therefore,

$$\frac{\gamma_A}{\gamma_B} = \frac{t_{1/2,B}}{t_{1/2,A}} = \frac{1000}{100} = 10.$$

For linear air damping, the damping coefficient is inversely proportional to the mass,

$$\gamma = \frac{b}{m},$$

where b is the drag constant. Since the two pendulums have identical geometry,

$$b_A = b_B.$$

Thus,

$$\frac{\gamma_A}{\gamma_B} = \frac{m_B}{m_A}.$$

Using $m_A = 9 \text{ kg}$,

$$m_B = 10 m_A = 90 \text{ kg}.$$

Therefore, $m_B = 90 \text{ kg}$.
